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Problem Statement - Queues are frequently used in computer programming, & a typical example is the simulation of a job queue by an operating system does not use priorities, then the jobs are processed in the order they enter the system. Write a C program for simulating a job queue, where func. to add a job & delete a job.

Objectives -

1) Understand the concept of queue & FIFO
2) Implement a linear queue using an array
3) Implement functions to add & delete jobs
4) Simulate processing of jobs in the order they enter the system
5) Analyze the time & space complexity of queue operation

Platform -
- 64-bit open source Linux or its derivatives
- Open source programming

Psudo Code -

START
front = rear = -1
Repeat:
Display menu

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If Add :
    If rear == MAX-1 → Overflow
    Else add job at rear
If Delete :
    If queue empty → Underflow
    Else ~~add~~ remove job from fro
If Display :
    Display jobs from front to r
If Exit :
    STOP
END

Time Complexity -

- Enqueue : O(1)
- Dequeue : O(1)
- Display : O(n)
- Space : O(n)

Conclusion -

The job queue simulation demonstrates t
FIFO principle of a linear queue. J
are added at the rear & processed
removed from the front in the same or
in which they enter the system.

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FAQs

1) What is queue?

It is a linear data structure where insertion is done at the rear & deletion at the front.

2) What is FIFO?

FIFO means First In First Out. The element inserted first is deleted first.

3) Why is a queue suitable for an OS job queue?

Jobs are processed in the same order in which they enter, which follows FIFO.

4) What is queue overflow?

Overflow occurs when we try to insert a job into a full queue.

5) What is queue underflow?

Underflow occurs when we try to delete a job from an empty queue.

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6) What are applications of queue?

OS job scheduling, printer queues, CP scheduling & customer service syste

7) What is circular queue?

It is a queue in which the last position connected back to the first position.

8) Why is circular queue preffered over linea

It makes better use of the available array by reusing free positions.

9) What principle does circular queue follow?

FIFO principle

10) What are the basic operations of circular queue?

Enqueue, Dequeue, Peek & Display

11) What is overflow & underflow?

Overflow → circular queue is completely full
Underflow → when we try to delete an element from an empty circular queue.

```c
#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1, rear = -1;

void addJob()
{
    int job;

    if (rear == MAX - 1)
    {
        printf("Queue Overflow!\n");
        return;
    }

    printf("Enter Job ID: ");
    scanf("%d", &job);

    if (front == -1)
        front = 0;

    rear++;
    queue[rear] = job;

    printf("Job added successfully.\n");
}

void deleteJob()
{
    if (front == -1 || front > rear)
    {
```

```c
void deleteJob()
{
    if (front == -1 || front > rear)
    {
        printf("Queue Underflow!\n");
        return;
    }

    printf("Job %d processed/deleted.\n",
queue[front]);
    front++;

    if (front > rear)
        front = rear = -1;
}

void displayJobs()
{
    int i;

    if (front == -1)
    {
        printf("Queue is empty.\n");
        return;
    }

    printf("Pending jobs: ");
    for (i = front; i <= rear; i++)
        printf("%d ", queue[i]);

    printf("\n");
}
```

```c
int main()
{
    int choice;

    do
    {
        printf("\n--- Job Queue Menu ---\n");
        printf("1. Add Job\n");
        printf("2. Delete/Process Job\n");
        printf("3. Display Pending Jobs\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                addJob();
                break;

            case 2:
                deleteJob();
                break;

            case 3:
                displayJobs();
                break;

            case 4:
                printf("Exiting program.\n");
                break;
```

```
        default:
            printf("Invalid choice!\n");
        }
    } while (choice != 4);

    return 0;
}
```